

Audio Controller User Manual (AMR-S24 Series)

V2.2



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คู่มือนี้จะอธิบายวิธีใช้ผลิตภัณฑ์และเรื่องความปลอดภัย

อ่านคู่มือนี้อย่างละเอียดและคำนึงถึงความปลอดภัยระหว่างการใช้งาน

หากมีคำถามหรือข้อสงสัยเพิ่มเติม อย่าลังเลที่จะติดต่อเรา
เราพร้อมให้ความช่วยเหลือทุกเมื่อ

All Mobile Robotics Co., Ltd.

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1Safety

1.1 Safety warning signs

The following safety warning signs are used to warn of potential personal injury hazards. Please be sure to follow all safety information with this sign to avoid possible injuries.



Warning

Violation of the contents indicated in this warning during operation may result in death or serious injury.



Note

Violation of the contents indicated in this precaution during operation may result in personal injury or damage to items.



Reminder

To enable you to use the product correctly during operation, you must abide by the relevant usage instructions.

1.2 Precautions for safe use

The purpose of the precautions mentioned here is to enable you to use the product safely and correctly, and to prevent problems before they occur, so as to avoid harm and injury to you and others. please

Please fully understand its contents before using this product.



Notice

Do not use this product in explosive gas environments, flammable gas environments, corrosive environments, places prone to water, or near combustible materials. Otherwise, it may cause product failure, fire, or injury.

To prevent the risk of fire and electric shock, use this product at rated voltage.

Please connect according to the connection diagram, otherwise it may cause fire.

Maintenance/inspection must be performed after cutting off the power supply, otherwise injuries may occur. Please do not disassemble or modify this product, otherwise it may cause injury or damage to the device. For internal inspection or repair, please contact our company.



Remind

Please do not use this product in an environment with strong radio waves and strong electromagnetic radiation. The speaker may be interfered. When you install each component, please follow our recommended tightening torque. Please do not use this product where the vibration exceeds the specified range. Risk of product malfunction and falling. When using it outdoors, be sure to insulate the junction of the wires to avoid malfunctions and current leakage.

2 Product Overview

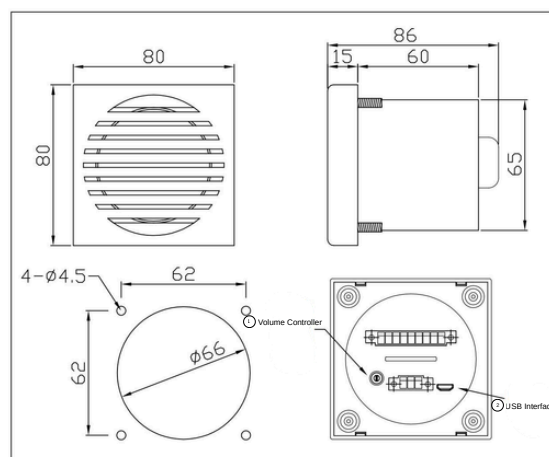
- Use a high-quality digital power amplifier to support WAV, MP3, and other audio format data.
- Uses stable and reliable NAND Flash memory chip with a capacity of 128M; external B-type USB interface allows free replacement of audio files.
- Supports I/O control, 8 bidirectional optocoupler isolated input control signals, and can control up to 255 channels of voice playback.
- Supports RS485, RS232, and CAN interface control, can customize communication protocols, and is compatible with PLC communication protocols of various brands.
- DC12V~36V wide voltage input.
- Supports up to 6 switching outputs and can control other equipment such as alarm lights.
- Onboard standard 3.5MM audio AUX, stereo output, can be connected to external speakers, sound columns, and other power amplifier equipment.
- Onboard audio output terminal for external passive speakers. Direct sound generation, is easy to use, and the maximum volume can reach more than 120 decibels, which is suitable for various noisy factory environments.
- On-board volume knob, you can freely adjust the volume according to the use environment.
- Standard guide rail or screw fixation, suitable for voice broadcasting in security alarms, advertising playback, AGV, robots, and automation equipment production lines.

3 product specifications

3.1 Specifications

Product specifications	AMR-S24-IO	AMR-S24-485	AMR-S24-CAN
Control method	I/O	RS485	CAN
Number of channels	255	255 (can be increased)	
Rated voltage	DC 24V		
Voltage support	DC 12V~36V		
Current	800mA		
Storage type	NAND Flash		
Capacity maximum	128MB		
Volume	100db (1000hz, 1m) (depending on the speaker) volume can be adjusted		
Default play	loop playback in the folder		
Rated power	5W		
External speaker	-		
Material	ABS		
Protection level	IP23		
Temperature	-20~55℃		
Ambient humidity	85%RH (no condensation)		
Weight	180g±10%		

3.2 Installation dimensions



4 Usage

4.1 Tag Definition



SPK-5-24-IO



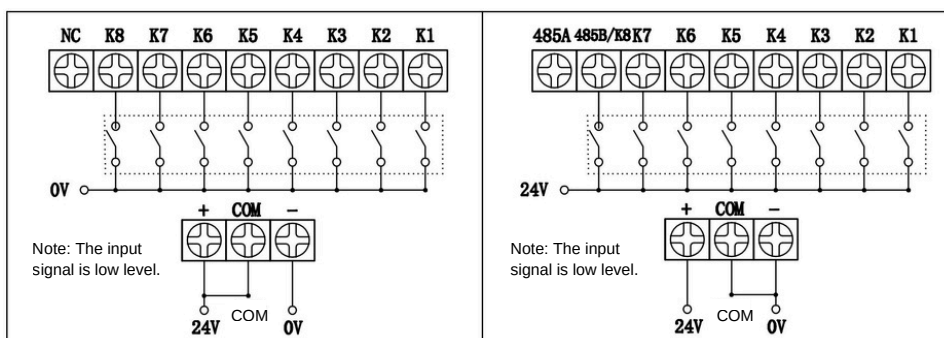
SPK-5-24-485



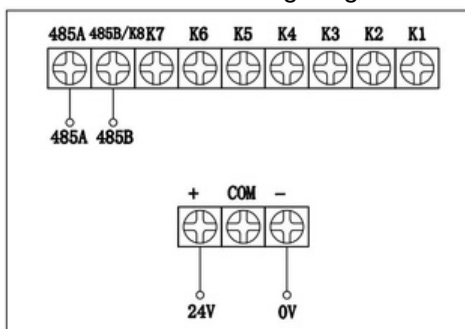
SPK-5-24-232 and SPK-5-24-CAN

4.2 Wiring Diagram

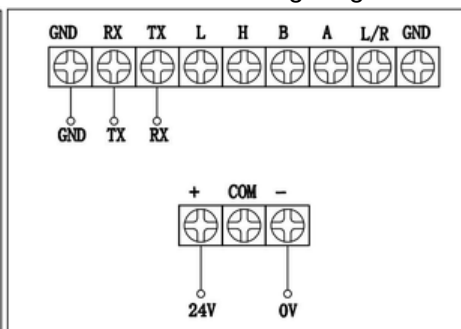
SPK-5-24-IO wiring diagram:



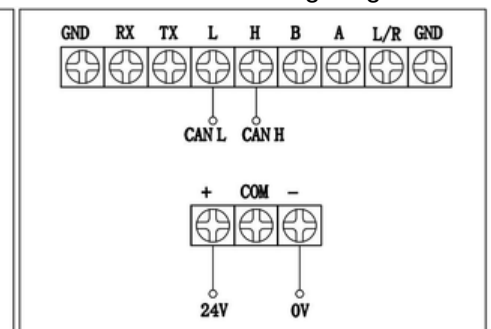
■ SPK-5-24-485 wiring diagram:



■ SPK-5-24-232 wiring diagram:



■ SPK-5-24-CAN wiring diagram:



4.3 Audio file download method

There is a Micro USB interface on the back of this product. When the product is powered off, use a USB data cable to connect it to a PC, and the PC will automatically generate a removable disk. Multiple folders can be created in the removable disk. The folder format is fixed to five digits "AW****". Each folder corresponds to an audio channel, i.e. AW001-AW255, corresponding to the 1st channel to the 255th channel respectively. Copy the prepared audio files to the corresponding folders. The audio format supports WAV and MP3 formats.



remind

- ☐ This method is only applicable to products with pre-stored audio files as the source type. After copying
- ☐ the data, please disconnect it correctly according to the Windows prompts to avoid damage to the storage card or data loss.

4.4 Volume Adjustment Methods

The product is equipped with a volume adjuster on the rear panel. Rotate the knob clockwise to increase the volume and counter-clockwise to decrease the volume.

For models AW-S24AF, AW-S24AT, and AW-S24AC, the volume can also be adjusted via software commands in addition to the manual volume adjuster.

4.5 Playback Control

The following examples are applicable when the voice announcer is set to a baud rate of 9600bps, 8 data bits, 1 stop bit, None parity, and a slave address of 1. (Applicable only to RS232 and RS485 communication interfaces).

4.5.1 AW-S24AG Control Method

The product is equipped with K1-K8 eight-channel bidirectional optocoupler isolated inputs. It utilizes binary coding rules to provide a total of 255 selectable channels Example of Folder Control Command (8-Bit Control)

The folder is controlled by sending ON/OFF signals to pins K1–K8.

ON = 1

OFF = 0

This is an 8-bit control signal.

Example 1: Play Folder AW001

K8	K7	K6	K5	K4	K3	K2	K1
0	0	0	0	0	0	0	1

Example 2: Play Folder AW003

K8	K7	K6	K5	K4	K3	K2	K1
0	0	0	0	0	0	1	1

Example 3: Play Folder AW050

K8	K7	K6	K5	K4	K3	K2	K1
0	0	1	1	0	0	1	0

Trigger Port	Playback Folder
K1	AW001
K2	AW002
K1, K2	AW003
K3	AW004
K4	AW008
K5	AW016
...	...

4.5.2 AMR-S24-485 playback method

The product adopts RS232 communication interface and supports Modbus RTU communication protocol and custom communication protocol. The default is custom communication protocol. See Chapter 6.1 for specific parameters.

AMR-S24BG/AMR-S24BT communication protocol.

□ Example: Use a custom communication protocol to play the audio in the folder "AW005" with a volume level of 28,

The host computer sends: 01 51 05 00 1C 49 02

The announcer returns: 01 52 05 00 1C 4A 02

4.5.3 AMR-S24-CAN playback method

The following example applies to the baud rate of the announcer being 125Kbps, the frame type being standard frame, the frame format being data frame, and the sending frame ID being 200.

The product uses a CAN communication interface and supports CAN 2.0A and CAN 2.0B communication protocols.

For specific parameters, see Chapter 6.2 AW-S24BC Communication Protocol.

□ Example: Play the audio in the folder "AW005" with a volume level of 28

The host computer sends: 01 51 05 1C 00 00 49 02

The announcer returns: 01 52 05 1C 00 00 4A 02



remind

- Before turning on the power, be sure to confirm that the wiring is correct.
- Do not use a data cable to connect to a PC during playback.
- The customized agreement shall prevail.

6 Communication protocol

6.1 AMR-S24-485 communication protocol

6.1.1 Custom protocol

This protocol playback mode defaults to loop playback in the folder. The playback mode and communication parameters can be customized according to customer requirements. The customized content of the customized protocol shall prevail.

■ Basic communication parameters

Communication method	Baud rate	Data bits	Stop bit	Check digit
RS232/RS485	9600bps	8 bits	1 bit	No check

■ The host computer sends data (length is 7 bytes)

0	1	2	3	4	5	6
first code	function code	Music folder number	spare	Volume	XOR values	end code
01	51	Data1	00	Data2	X0R	02

■ Explanation

● Data1 music folder number (1 byte, range 00~FF)

Data1 value	corresponds to the music folder number
00	Stop playing
01	AW001
02	AW002
0F	AW015
1F	AW031
...	...
FF	AW255

● Data2 volume (1 byte, range 00~1C, 29 levels in total)

Data2 value	corresponds to the music folder number
00	0
01	1
02	2
0F	15
...	...
1C	28

■ Announcer reply data

0	1	2	3	4	5	6
first code	function code	Music folder number	spare	Volume	XOR values	end code
01	52	Data1	00	Data2	X0R	02

■ Example

Play audio files from AW002 at volume level 28.

Host computer sends 015102001C4E02

0	1	2	3	4	5	6
first code	function code	Music folder number	spare	Volume	XOR values	end code
01	51	02	00	1C	4E	02

Host computer sends 015202001C4D02

0	1	2	3	4	5	6
first code	function code	Music folder number	spare	Volume	XOR values	end code
01	52	02	00	1C	4D	02

6.1.2 Modbus RTU protocol

This protocol playback mode defaults to loop playback in the folder. The playback mode and communication parameters can be customized according to customer requirements. The customized content of the customized protocol shall prevail.

■ Basic communication parameters

Protocol Modbus RTU	Baud rate	Data bits	Stop bit	Check digit	Device address	Function code
slave	9600bps	8 bits	1 bit	No check	01	03/06/10

■ Modbus register address table:

Register address (decimal)	Function	Value read and write (decimal)	permissions	Function Description
1	-	-	-	-
3	Audio folder number	0~255	Read and write	0: Stop playing 1: Play the AW001 folder 2: Play the AW002 folder 255: Play AW255 folder
2	-	-	-	-
4	Playback volume	0~28	Read and write	0: Quiet, 28: Highest
.....				

03 function code: read multiple register data

■ Example: The host computer reads the values of registers [3] and [4], that is, obtains the audio file played by the announcer and the playback volume. Host computer sends: 010300030002340B

Action	address	function code	Start high bit	Start low bit	High registers	low registers	CRC high	CRC low
Send	01	03	00	03	00	02	34	0B

□ 06 function code: write single register data

■ Example: The host computer writes 17 to register [3], that is, the audio file in AW017 is played.

Host computer sends: 01060003001739C4

Action	address	function code	total bytes	Start high bit	Start low bit	High registers	low registers	CRC high	CRC low
Send	01	03	04	00	FF	00	14	CA	0C

Announcer returns: 01060003001739C4

Action	address	function code	total bytes	Start high bit	Start low bit	High registers	low registers	CRC high	CRC low
Send	01	03	04	00	FF	00	14	CA	0C

□ 10 Function code: Write multiple register data ■

Example: Play the audio file in the file "AW255", the volume is 20, and the registers [3] and [4] are written with 255 and 20 respectively (hexadecimal is FF and 14 respectively).

Host computer sends: 01 10 00 03 00 02 04 00 FF 00 14 83 85

Action	address	function code	Start high bit	Start low bit	Register total number high bit	High registers	low registers	Register total	Register value high bit
Send	01	10	00	03	00	02	04	00	FF

Action	High registers	low registers	CRC high bit	CRC low bit
Send	00	14	83	85

Announcer returns: 011000030002B1C8

Action	address	function code	Start high bit	Start low bit	High registers	low registers	CRC high	CRC low
Send	01	10	00	03	00	02	B1	C8

6.2 AMR-S24-CAN communication protocol

This protocol playback mode defaults to loop playback in the folder. The playback mode and communication parameters can be customized according to customer requirements. The customized content of the customized protocol shall prevail.

■ Basic communication parameters

Communication specifications	baud rate	Frame type	frame format	Send frame ID	Feedback frame ID	Send cycle
CAN 2.0A/B	500Kbps	standard frame	Data Frame	200	201	100ms

■ Controller sends CAN frame (8 bytes in length)

0	1	2	3	4	5	6	7
first code	function code	Music folder number	Volume	Backup	Backup	XOR values	end code
01	51	Data1	Data2	00	00	X0R	02

■ Explanation

- Data1 music folder number (1 byte, range 00~FF)

Data1 value	corresponds to the music folder number
00	Stop playing
01	AW001
02	AW002
0F	AW015
1F	AW031
...	...
FF	AW255

- Data2 volume (1 byte, range 00~1C, 29 levels in total)

Data2 value	corresponds to the music folder number
00	0
01	1
02	2
0F	15
...	...
1C	28

- XOR exclusive OR value

The value obtained by XORing the previous six bytes.

■ Announcer feedback CAN frame (8 bytes in length)

0	1	2	3	4	5	6
first code	function code	Music folder number	Volume	Backup	Backup	XOR values
01	52	Data1	Data2	00	00	X0R

■ Example Play audio files in AW002 at volume level 28.

The controller sends 0151021C00004E02 frame ID200 and the announcer replies 0152021C00004D02 frame ID201

7 Troubleshooting

Fault phenomenon	Check and confirm	Treatment measures
Can't play	Is the power turned on normally?	Confirm whether the transmission signal cable is wired correctly
	Are the signal lines wired correctly?	Confirm whether the transmission signal cable is wired correctly
	Are the serial port and instructions correct?	Confirm whether the serial port is normal and whether the instructions are correct
	Is the USB cable disconnected?	Make sure the USB cable is disconnected, then power on again to test
	Is there a corresponding audio file in the memory?	Confirm audio files in memory
	Is the speaker on?	Make sure the speaker is connected properly
	Whether the volume is at the minimum state	Adjust volume
Play is different from specified voice message	Are the signal lines wired correctly?	Confirm whether the transmission signal cable is wired correctly
	Are the serial port instructions correct?	Confirm whether the instructions are correct
	Are the audio files in the memory configured correctly?	Confirm whether the audio files in the memory are correct
Storage not recognized	Is the power supply disconnected?	Make sure the power is disconnected, then reconnect the USB
	Is the USB data cable normal?	Check whether the USB data cable has data transmission function
	Are the player and computer USB ports normal?	Check whether the player and computer USB ports are normal

8 After-sales service

8.1 Warranty period

The voice controller provides a limited-time warranty. If the product cannot be used normally due to quality problems, design defects, etc. during the warranty period, we will provide free after-sales maintenance.

■Warranty period: 6 month from sale

8.2 Warranty scope

If the product is within the warranty period and falls within the warranty conditions, we will repair or replace it free of charge.

- The warranty and after-sales service of this product are limited to mainland Thailand;
- The product cannot be used normally due to unpacking due to transportation;
- The product itself cannot be used normally due to damage to its components;
- The product cannot be used normally due to design defects.

8.3 Scope of disclaimer

Please pay attention to the following conditions when using the product. We will not provide free after-sales and warranty.

- Failure to install the product correctly in accordance with the instructions, causing product damage;
- Using the product in unsuitable environments and conditions, causing product damage;
- Failure to operate the product according to the specifications of the product instructions, causing damage to the product;
- Unauthorized use of the product without the permission of the company Disassemble or repair the product;
- Product damage caused by irresistible external forces such as natural disasters and fires.

8.4 Technical support

บริษัท ออล โมบายล์ โรโบติกส์ จำกัด

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